



Salt Diffusion Up the Mississippi River



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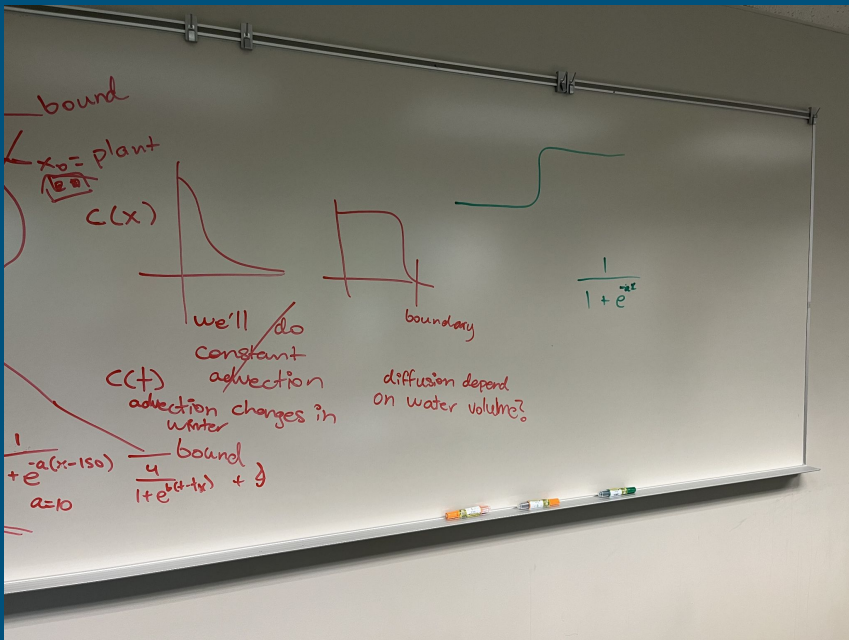


Background

- The Mississippi water flow is down
- New Orleans water level has decreased
 - Worries of salt reaching the water processing plant
- Good News: New Orleans is no longer in danger of salt water intrusion

Model Assumptions

- Flow does not change based on position
- Salt level in the ocean is constant
- We simplify to 1 position dimension
 - Flow and salination do not change based on height or width of the river
- Diffusion is constant



Our Model

- Normal water flow: $c = 4.8$ km/h
- Salt level in the ocean: $k = 36.5$
- Diffusion: $\alpha = 30$
 - Chosen to create a reasonable curve
- $a = 1, b = 1$
- Water processing plant is about 150 km away from the ocean
- Equation H is used to help keep the ocean's salt concentration constant
- So the forcing term stays on the ocean w/o dumping salt unnecessarily
- Boundary Conditions:
 - $u(-100, t) = 0$
 - $u(300, t) = k = 36.5$

$$u(t = 0, x) = k * H$$

$$H(x) = \frac{1}{1 + e^{-a(x-150)}}$$

$$c(t) = \frac{2}{1 + e^{-b(t-20)}} + 2.8$$

$$u_t = 2u_{xx} - c(t)u_x + H(x)(k - u)$$

Live Demo

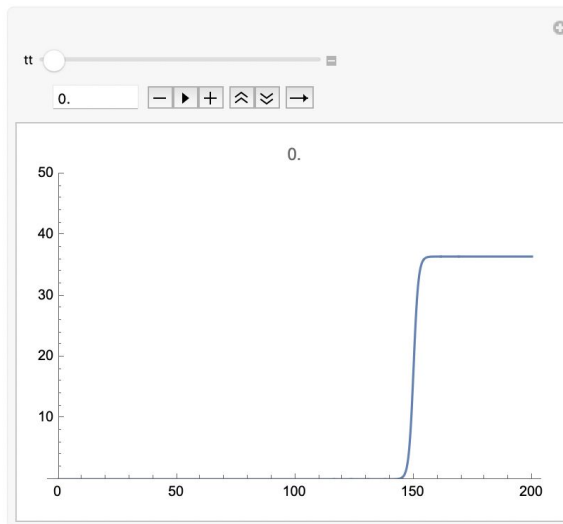
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in[85]:= Clear[u, x, t]

finit[x_] :=  $\frac{36.5}{1 + e^{-(x-150)}}$ ;

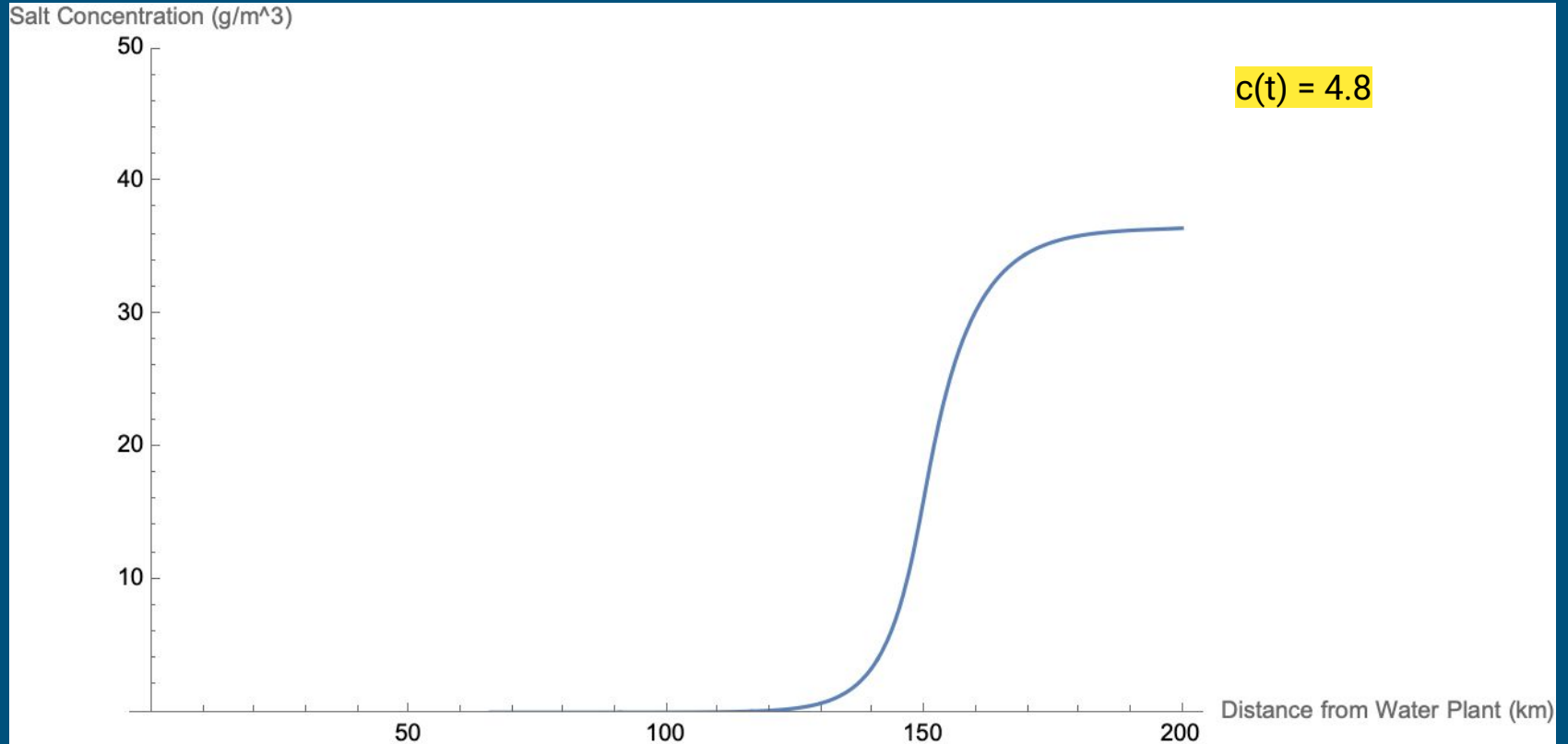
c[t_] :=  $\frac{2}{1 + e^{t-40}}$  + 0.8

pde = D[u[x, t], t] + c[t] * D[u[x, t], x] == 30 D[u[x, t], x, x] +  $\frac{1}{1 + e^{-(x-150)}}$  * (36.5 - u[x, t]);

lbc = u[-100, t] == 0;
rbc = u[300, t] == 36.5;
soldiff = NDSolve[{pde, lbc, rbc, u[x, 0] == finit[x]}, u[x, t], {x, -100, 300}, {t, 0, 200}, MaxStepFraction -> 1/100];
Manipulate[Plot[u[x, t] /. soldiff /. {t -> tt}, {x, 0, 200}, PlotRange -> {0, 50}, PlotLabel -> tt], {tt, 0, 200}]
area[tt_] := NIntegrate[u[x, t] /. soldiff /. {t -> tt}, {x, -100, 300}]
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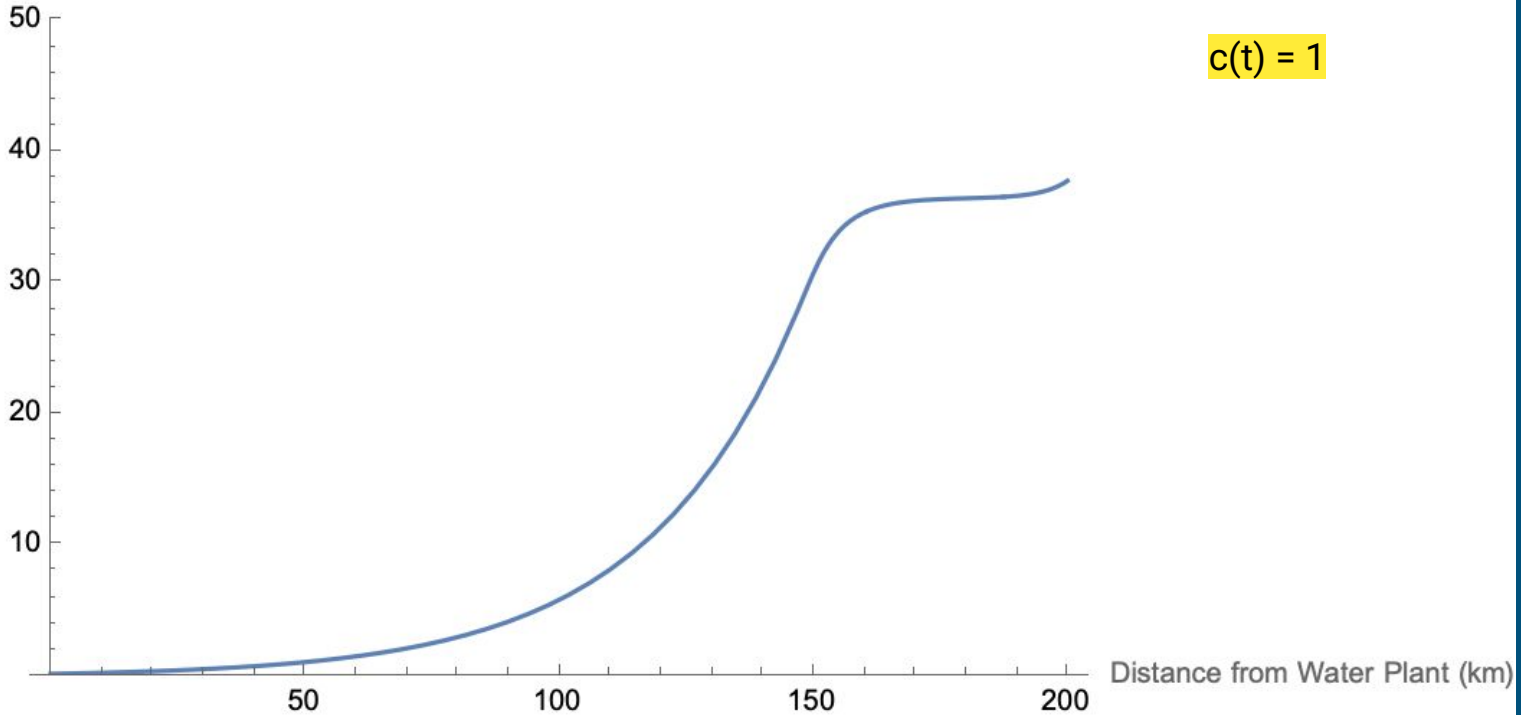
Steady States



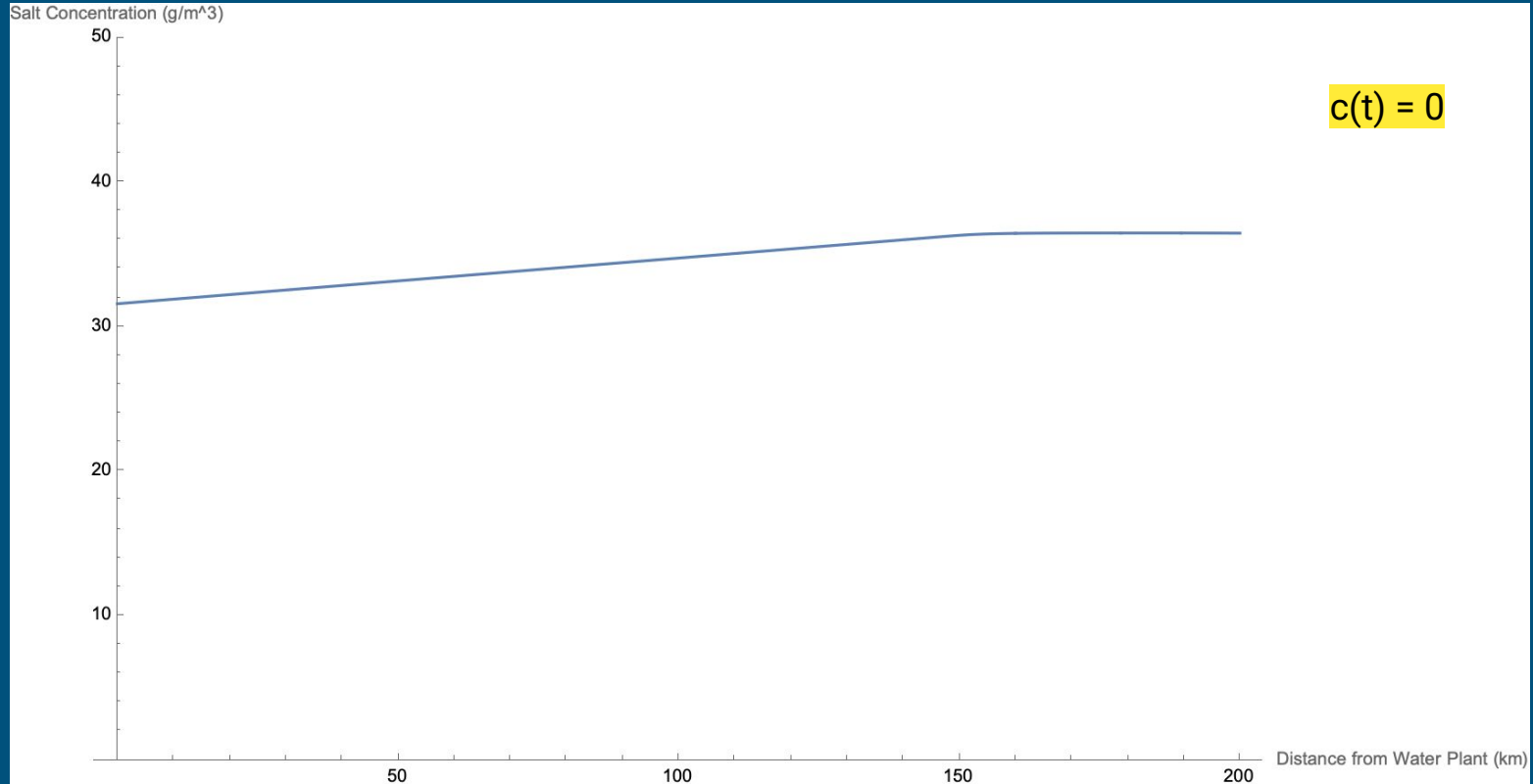
Steady States

Salt Concentration (g/m³)

$$c(t) = 1$$



Steady States



Model Interpretation

- Both the live demo and steady states show salt approaching the water processing plant based on advection
- 4.8 down to 1 is ~80% decrease in water flow and shows significant difference in salt concentration levels
- Water processing plant may receive salt if water flow decreases around 80% or greater depending on diffusion rates
- Climate change impacting water levels has ripple effects

Further research

- Getting more precise data for New Orleans
 - Flow rate
 - Facility distance
 - Change in flow for this season
 - Finding an actual water diffusion coefficient
- Fitting our model to actual data
 - How accurately does our model predict salt levels in New Orleans?
- Understanding what models New Orleans is using
 - How does our model compare to theirs?

Resources:

- Mississippi River Speed: “Mississippi River Facts.” *National Parks Service*, U.S. Department of the Interior, www.nps.gov/miss/riverfacts.htm#:~:text=At%20New%20Orleans%20the%20river,reach%20the%20Gulf%20of%20Mexico. Accessed 19 Oct. 2023.
- “New Orleans No Longer under Imminent Threat of Saltwater Intrusion into Their Drinking Water Supply: Latest Weather Clips.” FOX Weather, Fox Weather, 13 Oct. 2023, www.foxweather.com/watch/play-6b29d3ab1000b67.
- Salt Concentration in the Gulf of Mexico: de Lanza Espino, Guadalupe, and Juan Carlos Gomez Rojas. *PHYSICAL AND CHEMICAL CHARACTERISTICS OF THE GULF OF MEXICO*, www.hartheresearch.org/sites/default/files/inline-files/3.pdf. Accessed 19 Oct. 2023.
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- Distance from New Orleans to Gulf of Mexico along the Mississippi River: “The Mississippi River in New Orleans.” *In New Orleans*, www.neworleans.com/things-to-do/recreation/the-mississippi-river-in-new-orleans/. Accessed 19 Oct. 2023.